GRAVITYSORT · PROBLEM STATEMENT

GravitySort problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Kit 1.8 pipeline: users-and-needs (Table 2) and stakeholders (Table 3) tables added for GVS-URS-001; constraint cost updated to the estimated USD 594 (GVS-CAL-001 v0.9); requirements now GVS-PRS-001
0.7	2026-10-02	Amish Chadha	Site water and first partner as decided on 2026-10-02 (GVS-DEC-001)
0.6	2026-10-01	Amish Chadha	Budget worded as a value-engineering target; cost of the constructable design (GVS-DDR-003)
0.5	2026-09-26	Amish Chadha	Budget approved by Amish ($455)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Budget $450; borax smelting step and partner screening criterion decided
0.3	2026-09-25	Amish Chadha	TRL 3 update. Cost and power constraints from GVS-CAL-001; water pumping gap; partner screening criterion and borax smelting step adopted for TRL 3 under Amish's

			2026-09-25 instruction (GVS-DDR-001), open for his review
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, out of scope, prior work with sources)
0.1	2026-09-25	Amish Chadha	Initial scaffold

GravitySort problem statement

Artisanal and small-scale gold miners use mercury to catch fine gold because it is cheap, needs no power and works well enough, and the mercury ends up in their bodies, their homes and their rivers. Mercury-free gravity equipment exists, but the machines that catch fine gold (centrifugal concentrators and shaking tables) are costly and mostly imported, and cheap sluices lose much of the fine gold. Design with, not for: requirements must come from co-design sessions and field trials with the intended users through a local partner.

The problem

Artisanal and small-scale gold mining (ASGM) is the largest source of human-made mercury emissions to air, at about 838 t in 2015, or 37.7 % of the global total (US EPA summary of the UNEP Global Mercury Assessment 2018). UNEP estimates that 10 to 15 million people work in the sector, including 4 to 5 million women and children (UNEP Global Mercury Partnership).

Mercury is used because it solves a real processing problem. Much of the gold in milled ore is fine (below about 0.1 mm), and fine gold is hard to hold on a sluice. Miners add mercury, which wets gold and forms an amalgam that is easy to collect, then burn off the mercury over an open fire to leave sponge gold. The two worst practices are whole-ore amalgamation, where mercury is mixed with all of the ore rather than a small concentrate, and open burning of amalgam. Both are named for elimination in Annex C of the Minamata Convention on Mercury (Minamata Convention guidance on national action plans).

Mercury is also not as good as it looks. In Colombian processing centers where whole ore is amalgamated in small ball mills, gold recovery is typically about 30 %, and a study of simple gravity and flotation equipment built from local materials found that concentration can be more profitable when recovery is higher than that (Veiga et al., *Journal of Cleaner Production* 205, 2018). The obstacle is less the physics than access: gravity equipment that catches fine gold is "generally more expensive" and "requires some experience to operate" (US EPA, ASGM without mercury).

GravitySort addresses the equipment gap: an open, garage-buildable centrifugal concentrator and small shaking table, powered by pedals or a small electric motor, that together make a clean concentrate small enough to smelt directly with borax, so no mercury is needed at any step.

Users and context

Table 1. Intended users. Proposed for review; the real list must come from co-design sessions.

User	Need	Context
Small mining group (3 to 10 people)	Recover more of the gold in the ore they already mine, without mercury, at a cost they can recover in weeks	Hard-rock or alluvial pits; ore milled by hammer mill or ball mill to below about 1 to 2 mm
Processing center (entable, plant) operator	Replace whole-ore amalgamation with a concentrate step to meet new rules and keep customers	Mills ore for many miners on a fee basis; has mains or generator power
Women processors	A safer role than panning with mercury at home or at the river	Often handle panning, washing and amalgam burning today
Local workshop or welder	Build, repair and sell the machine with local tools and materials	Welding set, drill press, hand tools; no lathe in many cases
NGO, cooperative or extension officer	Demonstrate mercury-free processing with miners and measure recovery	Programmes under national ASGM action plans; planetGOLD and similar

Table 2. Users and needs (kit 1.8). Proposed for review with the partner; the source of each user requirement in GVS-URS-001.

User type	Goals and needs	Typical use scenario	Pain points and challenges
Small mining group (3 to 10 people)	More gold from the same ore without mercury; equipment they can pay back in weeks; keep up with about 1.5 t of milled ore a day	Feeds milled ore by shovel and water through a shift, pedalling in turns; flushes the bowl every 2 h; cleans up on the table at the end of the day	Fine gold lost on sluices; no power at the pit; little and silty water; theft of concentrate; equipment imported, costly and hard to repair
Processing center (entable) operator	Replace whole-ore amalgamation with a concentrate step to meet the law and keep customers; run on mains or generator power	Runs the machine for several miners on a fee basis, with the motor	Customers expect fast turnaround; new rules on mercury; mixed ores from many pits
Women processors	A safer role than panning with mercury at home or at the river; a share of the clean-up work they already do	Clean-up on the table and smelting preparation; handling the locked concentrate	Exposure to mercury vapour and arsenic-bearing concentrate today; heavy, stooped panning
Local workshop or welder	Build, repair and sell the machine with local tools and materials	Welds the frame, casts or buys the liner, fits bicycle parts and bearings	No lathe in many shops; specialist parts not sold locally
NGO, cooperative or extension officer	Demonstrate mercury-free processing and measure recovery with miners	Runs demonstrations and trials with a partner cooperative under a national action plan	Needs equipment that works with local feed and a way to show recovery

Table 3. Stakeholders (kit 1.8).

Stakeholder group	Role or interest	Engagement needed
Miners' cooperative in Colombia, introduced through the Alliance for Responsible Mining (first candidate, decided by Amish on 2026-10-02)	Co-design partner and first field site	Co-design sessions on the user requirements; field trial at TRL 5 (on hold)
National mining and environment authorities (for example Colombia, where mercury in mining is banned under Law 1658 of 2013)	Enforce mercury bans and national action plans	Check how a mercury-free concentrator fits the rules the partner works under (GVS-REG-001)
planetGOLD and similar programmes	Fund and promote mercury-free ASGM	Share results; possible route to finance and spread
Gold buyers and refiners	Want traceable, mercury-free gold	Interest in where the gold comes from; no direct design input
Local communities downstream	Rivers and drinking water affected by mercury and silt	Settling pond practice and no mercury at any step
Helpful Engineering network and Design Molecule	Open design, documentation and partner introductions	Keep the design open (CERN-OHL-S) and documented

Operating environment

- **Feed:** ore milled to below 2 mm, as a slurry of about 25 to 35 % solids by mass; gold grades from about 1 to 20 g/t; heavy minerals such as magnetite, pyrite and hematite in the feed.
- **Sites:** remote pits and camps, often in tropical forest or savanna, sometimes at altitude; dust, mud, rain and heat (15 to 40 °C).
- **Water:** often scarce or silt-laden; recirculation through a settling pond is good practice and increasingly required.
- **Power:** frequently none, or a small generator shared with the mill.
- **Security:** gold concentrate is valuable and theft is a real concern, so concentrate must be easy to lock away.
- **Supply chain:** steel tube, sheet metal, bicycle parts, bearings, V-belts, HDPE drums and hoses are sold in most mining towns; specialist parts are not.

Constraints

- Garage-buildable prototype. Value-engineering target: USD 455 in parts, a hypothetical control target rather than a limit (Amish, 2026-10-01; raised from $350 to $450 on 2026-09-25 and to $455 on 2026-09-26, GVS-DDR-002). Estimated cost of the constructable design: USD 594 (USD 139 over the target; GVS-CAL-001 v0.9).
- No mercury in any step, and no chemicals beyond water and, at the smelting step outside this machine, borax flux.
- Built with welding, drilling and hand tools; parts that normally need a lathe (bowl, spindle) must have a no-lathe route.
- Runs without grid power: pedal drive as the baseline, with an optional motor through MotionCore on a pack of the user's choice. The water supply (about 1.19 m3/h) comes by gravity from upstream as the site rule for the first field trial, or from a bought treadle or hand pump or a 12 V pump where that is impossible (decided by Amish, 2026-10-02, GVS-DEC-001).
- Carried to site in parts by two people and assembled with hand tools.

Out of scope

- Milling and crushing (a separate machine; GravitySort starts from milled ore).
- Cyanide leaching, flotation and any chemical processing.
- Smelting equipment. Direct smelting with borax is the recommended final step (decided by Amish, 2026-09-25: go with recommendation; GVS-DDR-002) and the concentrate is sized for it, but the furnace is not part of this design.
- Tailings dams and water treatment beyond a settling pond.
- Legal status, licensing and the gold trade. These decide whether any equipment is adopted and must be covered by the partner organization.

Prior work

- **Commercial centrifugal concentrators** (Knelson, Falcon and similar) use a spinning riffled bowl, often fluidized with water, to hold dense particles at tens of times gravity. They are effective on fine gold but costly, and small units run batch cycles of about 0.5 to 2 h (US EPA).
- **Shaking tables** give high-grade concentrates but are "relatively expensive and require some experience to operate" (US EPA).
- **CICAN project, Colombia.** A chain mill, shaking table and flotation cell were built from junkyard and workshop materials for micro-miners processing under 2 t per day, with capital costs of US$2,600 to US$10,500 (Veiga et al., 2018). GravitySort aims below the bottom of that range for the concentration step alone.
- **Direct smelting with borax (Benguet method).** Small-scale miners in Benguet, Philippines, smelt gravity concentrates with borax instead of amalgamating them; the Benguet federation reports that its 15,000 members use the method (Pure Earth; Appel and Na-Oy, *Journal of Health and Pollution*, 2012). The method needs a small, clean concentrate; the US EPA cites about 50 to 100 g (US EPA).
- **planetGOLD** runs mercury-free ASGM programmes, including equipment and finance, in many countries (planetGOLD).

The gap GravitySort targets is an open design that combines the fine-gold capture of a centrifuge with the upgrading of a table, at a parts cost a small group can afford, and that runs on pedal power.

Open questions

- [ ] What gold particle size distribution do partner sites actually have, and how much of the gold is locked in sulfides that no gravity method will recover?
- [ ] Is pedal power acceptable for a full shift, or will users treat the motor as the default?
- [ ] How much water is available per site, and is recirculation practical?
- [ ] Which country's national action plan and partner offers the best first field site? The screening criterion (GVS-DDR-001 item 8a; decided by Amish, 2026-09-25: go with recommendation) is a country with a mercury ban in force, such as Colombia, or a Minamata action plan with a planetGOLD programme; the partner itself was decided by Amish on 2026-10-02 (GVS-DEC-001): Colombia, with a miners' cooperative introduced through the Alliance for Responsible Mining in Medellin as the first candidate to approach.
- [ ] How do sites without power get about 1.2 m3/h of water to the header tank? Gravity supply from upstream is the site rule for the first trial, with a treadle, hand or 12 V pump as the fallback (GVS-DEC-001); to be checked at the partner's sites.

User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise the user and product requirements (GVS-URS-001, GVS-PRS-001) from findings before freezing the design